

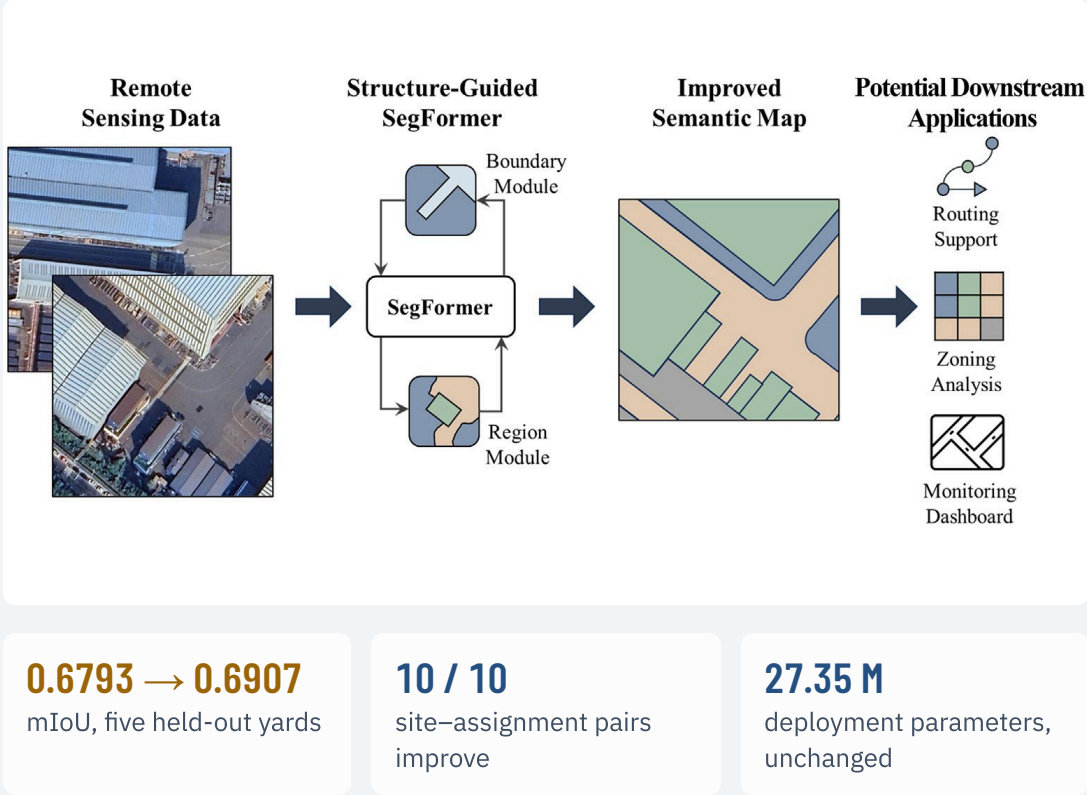
# Mapping roads, yards and buildings in shipyards the model has never seen

Structure-guided cross-site segmentation:  
boundary and region constraints used in  
training only, deployment model unchanged

*Structure-guided cross-site semantic segmentation of  
shipyard remote-sensing imagery*

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*Ocean Engineering* 368 (2026) 128269 · doi:10.1016/j.oceaneng.2026.128269

Figure: paper Fig. 1 (imagery Google, Airbus). \* Corresponding authors.



# The paper in one minute

1

## The task

Split shipyard imagery into background, road, yard open area and production building. In another yard, layout and appearance change: class transitions blur and large regions fragment.

2

## The method

Two training constraints on SegFormer-B2: a boundary constraint derived from the label morphology, and region consistency through eight learnable prototypes. Both are removed at inference.

3

## The results

Five held-out yards × three seeds: mIoU 0.6793 → **0.6907**, boundary F1 0.7190 → 0.7393, fragmentation 0.0775 → 0.0631. All 10 site-assignment comparisons improve.

4

## The cost

The deployed model is the baseline: 27.35 M parameters, 92.1 GFLOPs. Only training takes slightly longer: 32.20 → 34.18 s per epoch.

**Boundaries fix the transitions, regions fix the interiors: complementary, and free at deployment.**

# The order of this briefing (click to jump)

1

## Background and difficulty

What a shipyard semantic map is for; why a new yard is hard

p. 4

2

## Data and protocol

Eight yards, 988 patches; five held out in turn

p. 6

3

## Method

Boundary and region-consistency constraints

p. 8

4

## Main results

Overlap, classes, boundaries and fragments, per site, longer training

p. 11

5

## Mechanism and engineering value

Prototypes learn; free deployment; road connectivity

p. 16

6

## Qualitative results

Five yards compared; local gains and residual errors

p. 19

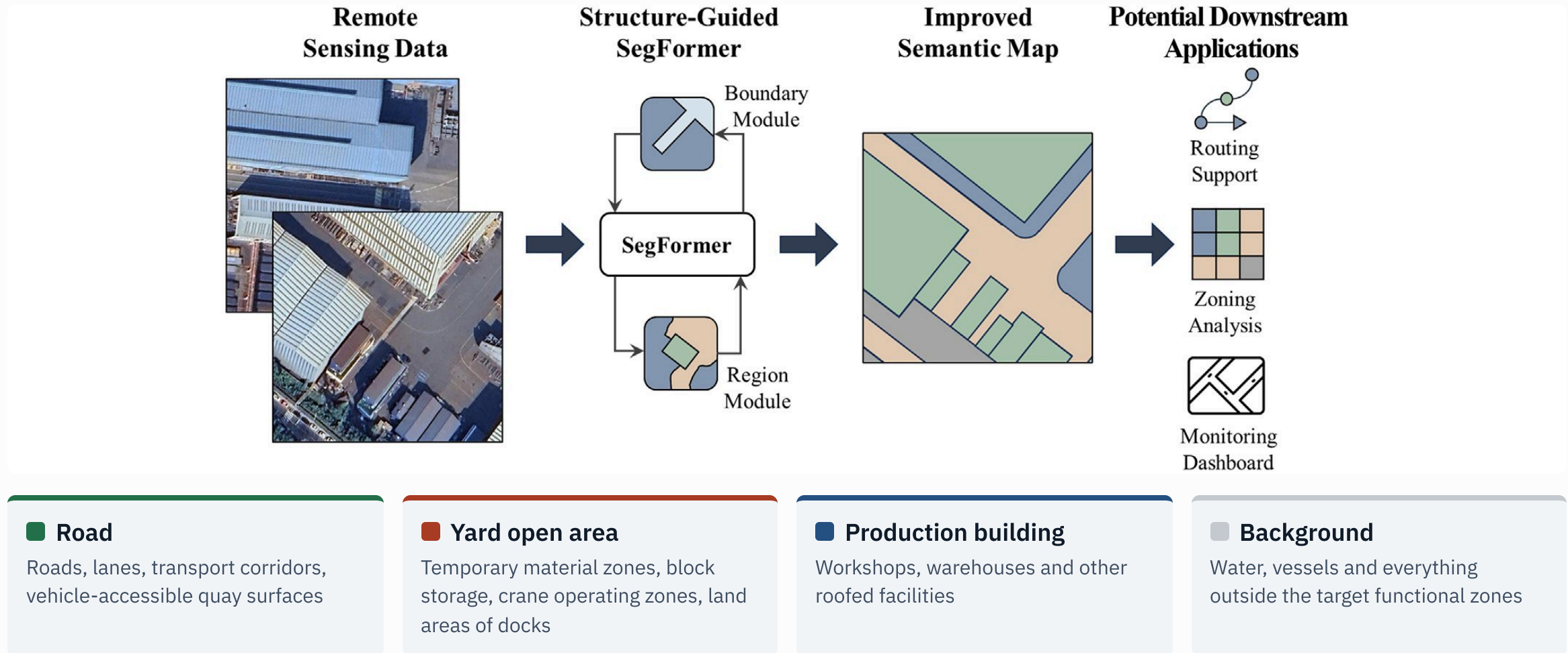
7

## Conclusions

Contributions and place in the thesis

p. 21

# A shipyard semantic map turns imagery into a spatial layer for digital twins and transport planning



Sources: paper Fig. 1 (imagery Google, Airbus) and the class definitions of §3.2. The map can feed route-network extraction, zoning, facility localisation and spatial indexing; those downstream modules are outside this paper.

# The difficulty: in another yard, layout, facilities, roads and appearance all change

1

## Scene-level distribution shift

Building density, road geometry, yard texture and organisation differ from yard to yard

2

## Weak boundaries between functional regions

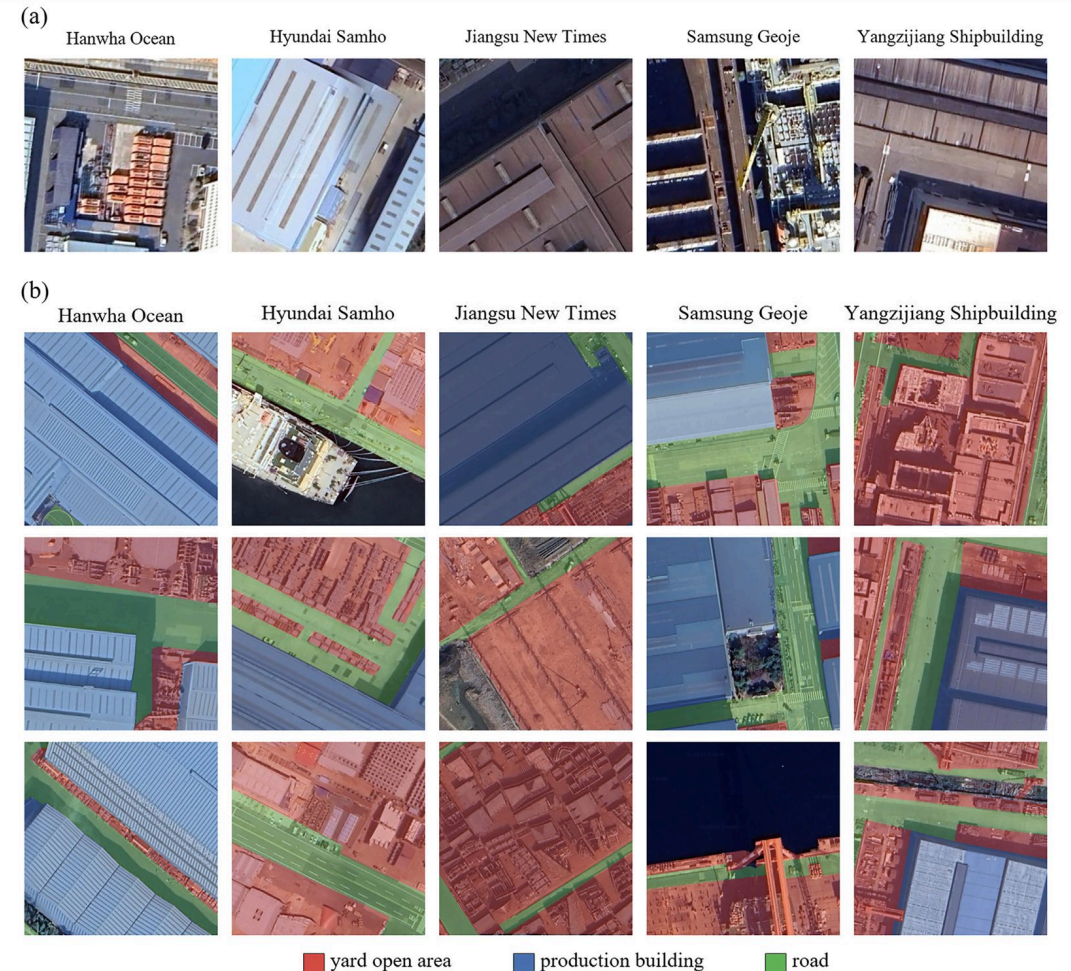
Road–yard and building edges are where errors concentrate

3

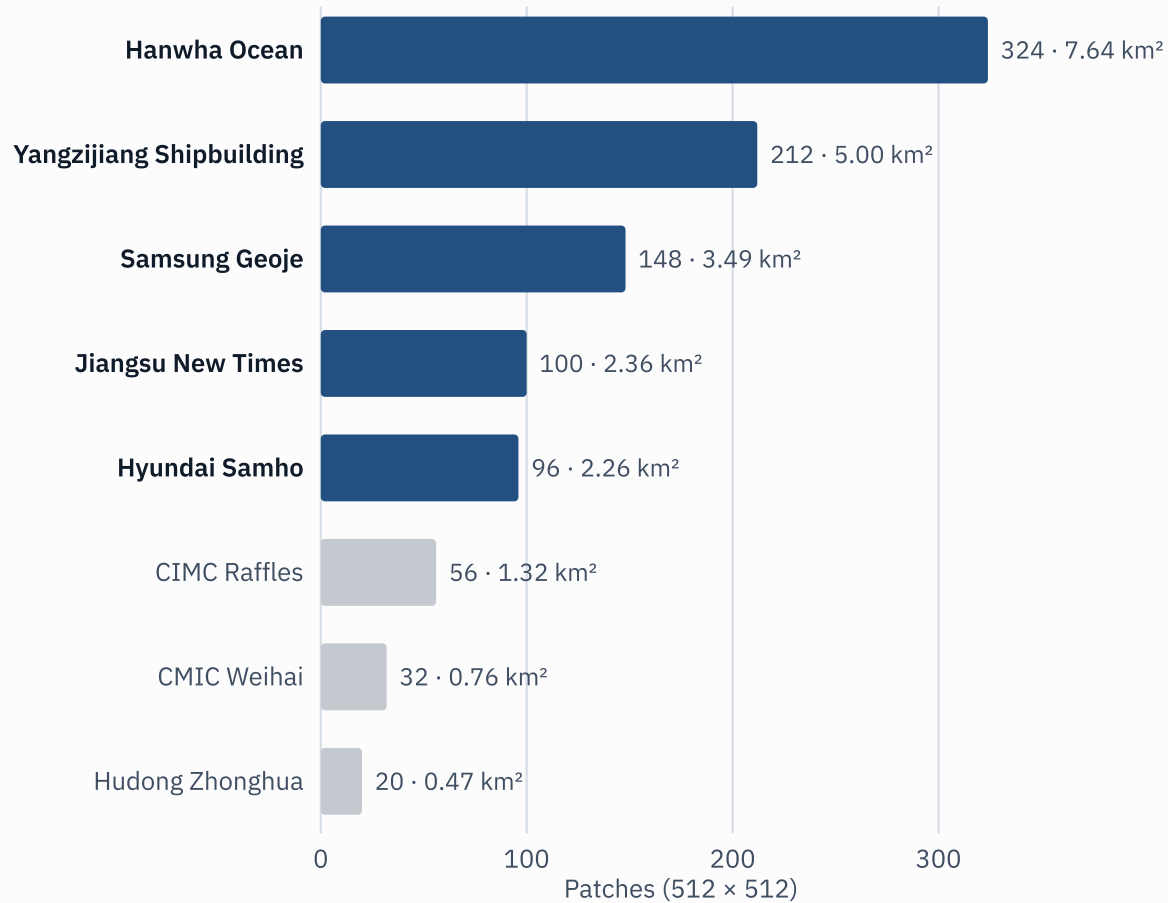
## Fragmented interiors

Uniform yards and roofs are predicted as scattered small pieces

**No image of the target yard is used for training or adaptation:  
this is domain generalisation, not adaptation.**



# Dataset: eight yards, 988 patches, about 23.3 km<sup>2</sup>, four roughly balanced classes



■ Held-out test yard (in turn) ■ Training only

Pixel share of the four classes



## 247 → 988

1024 × 1024 source images cut into four 512 × 512 quadrants; about 0.30 m/pixel; exported from Google Earth Pro in 2025

## 0.921 mIoU · 0.887 BF1

Three independent annotators with review and adjudication; pairwise agreement on a 99-patch QC subset (10.02%)

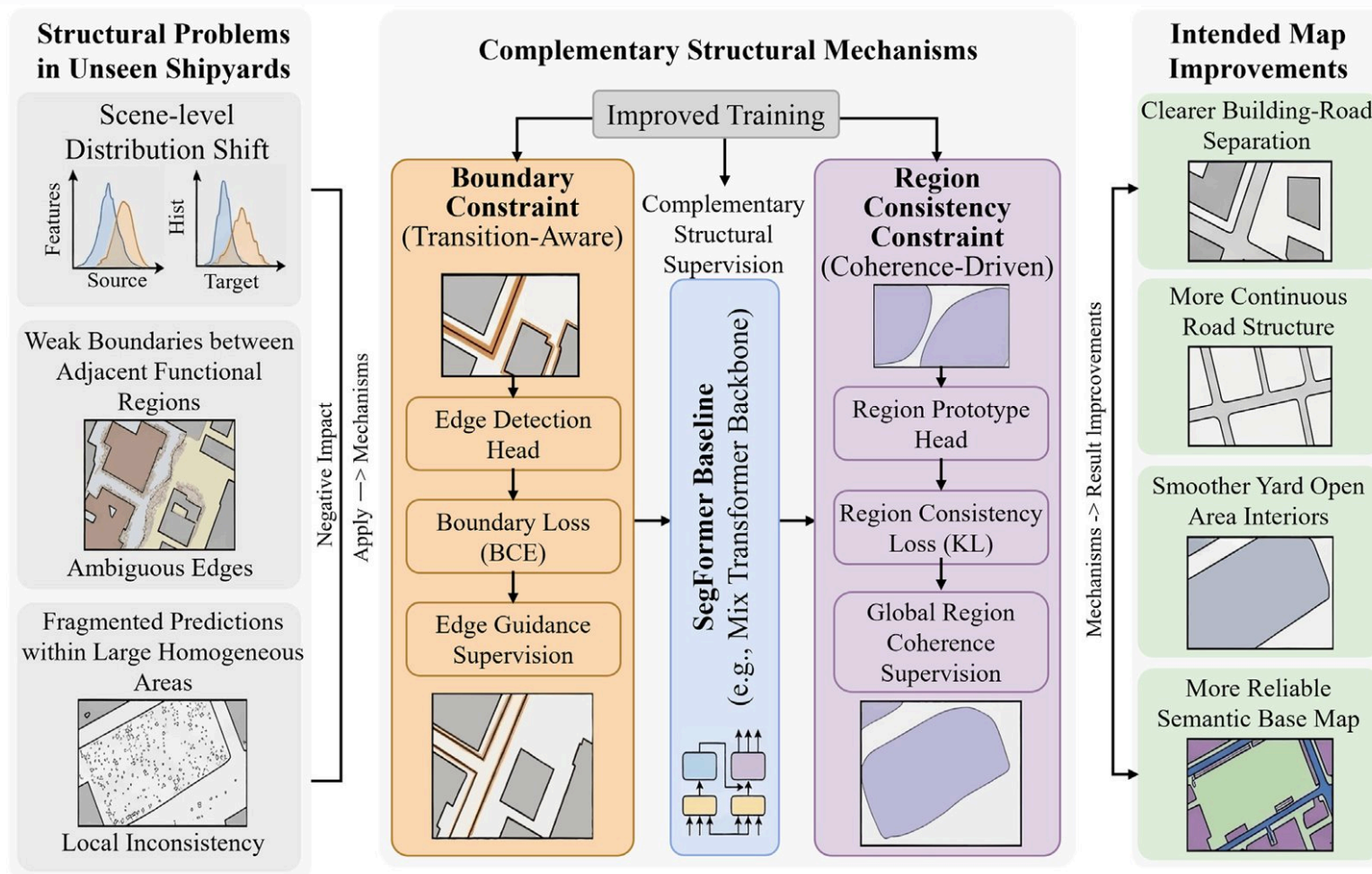
# Protocol: five yards held out in turn, with two validation-yard assignments

| TEST YARD                | VALIDATION YARD          | TRAIN / VAL / TEST PATCHES |
|--------------------------|--------------------------|----------------------------|
| Hanwha Ocean             | Samsung Geoje            | 516 / 148 / 324            |
| Hyundai Samho            | Hanwha Ocean             | 568 / 324 / 96             |
| Jiangsu New Times        | Hyundai Samho            | 792 / 96 / 100             |
| Samsung Geoje            | Yangzijiang Shipbuilding | 628 / 212 / 148            |
| Yangzijiang Shipbuilding | Jiangsu New Times        | 676 / 100 / 212            |

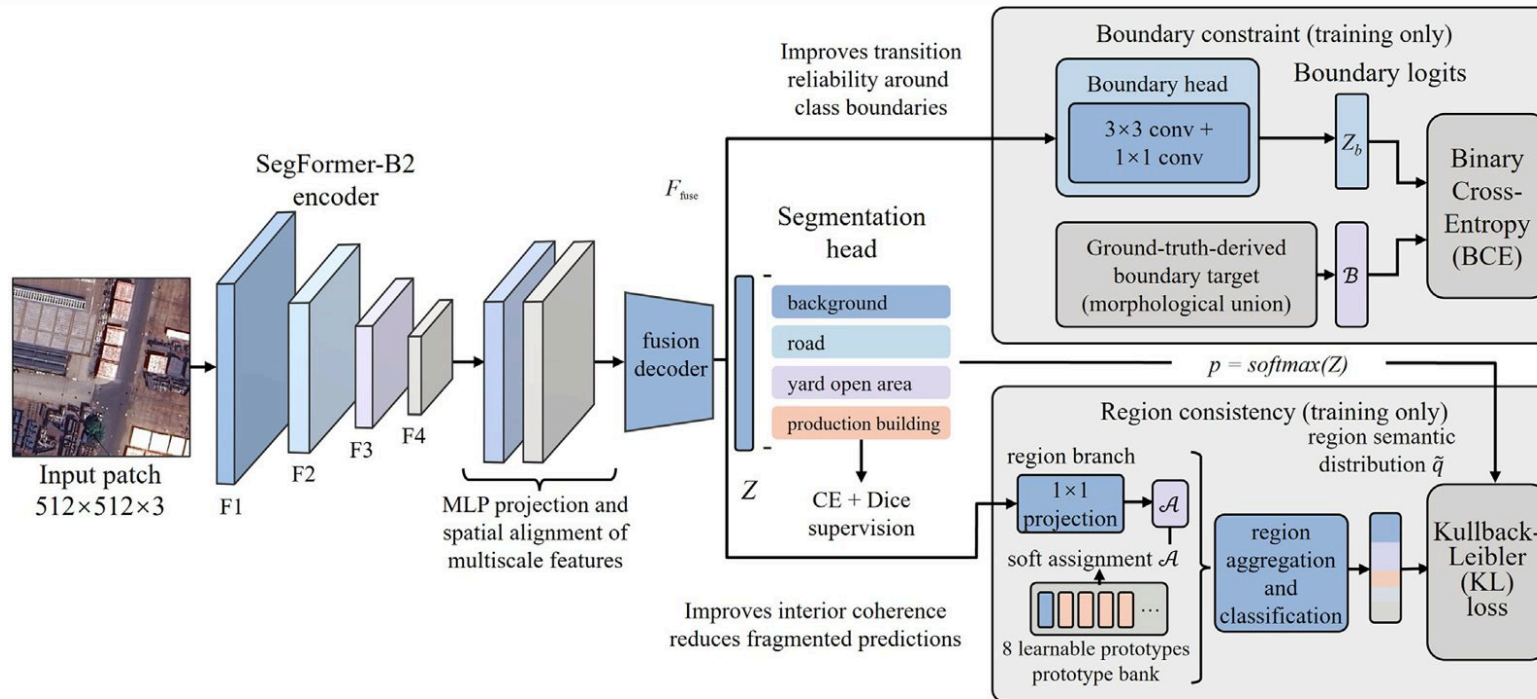
The table shows assignment B, used for the main comparison: one test yard, one validation yard and six training yards per fold.

- **Leave one shipyard out:** each of five well-sampled yards is the test yard in turn; no target-yard image is used for training or adaptation
- **Two validation assignments, A and B:** fixed before training, each a five-cycle; every test yard has a different validation partner in A and B
- **Three seeds** (3407, 5519, 7823) → 15 runs per method and assignment; checkpoints chosen by validation mIoU
- **Two budgets:** a controlled 30-epoch comparison (same-backbone ablation) and a 120-epoch external comparison

# The idea: a boundary constraint for transitions, a region-consistency constraint for interiors



# Architecture: both auxiliary branches exist only in training and are removed for inference



Joint training objective

$$\mathcal{L} = \mathcal{L}_{ce} + \lambda_{dice} \mathcal{L}_{dice} + \lambda_b \mathcal{L}_{boundary} + \lambda_r \mathcal{L}_{region}$$

$$\mathcal{L} = \mathcal{L}_{ce} + \lambda_{dice} \mathcal{L}_{dice} + \lambda_b \mathcal{L}_{boundary} + \lambda_r \mathcal{L}_{region} \quad \cdot \quad \lambda_{dice} = 1.0, \lambda_b = 0.5, \lambda_r = 0.1$$

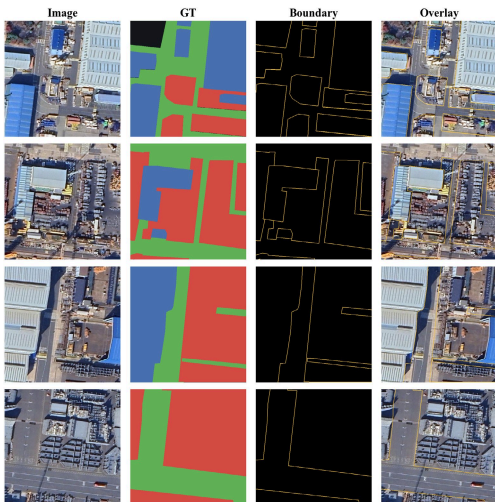
Three branches share the fused decoder features  $F_{fuse}$ : segmentation head (CE + Dice), boundary head (BCE), region-consistency branch (KL). The deployment graph remains SegFormer-B2.

# How the two constraints are built: no extra labels, no change at inference

## Boundary constraint: transitions

$$B = \max_c \text{clip}(\text{Dilate}(M_c) - \text{Erode}(M_c), 0, 1) \quad (3 \times 3)$$

- Boundary targets come straight from the label's morphological gradient, with no dedicated boundary labels; an ideal straight transition gives a band of about two pixels (0.6 m)
- A light boundary head ( $3 \times 3$  conv  $256 \rightarrow 128$ , BN, ReLU,  $1 \times 1$  conv) under BCE



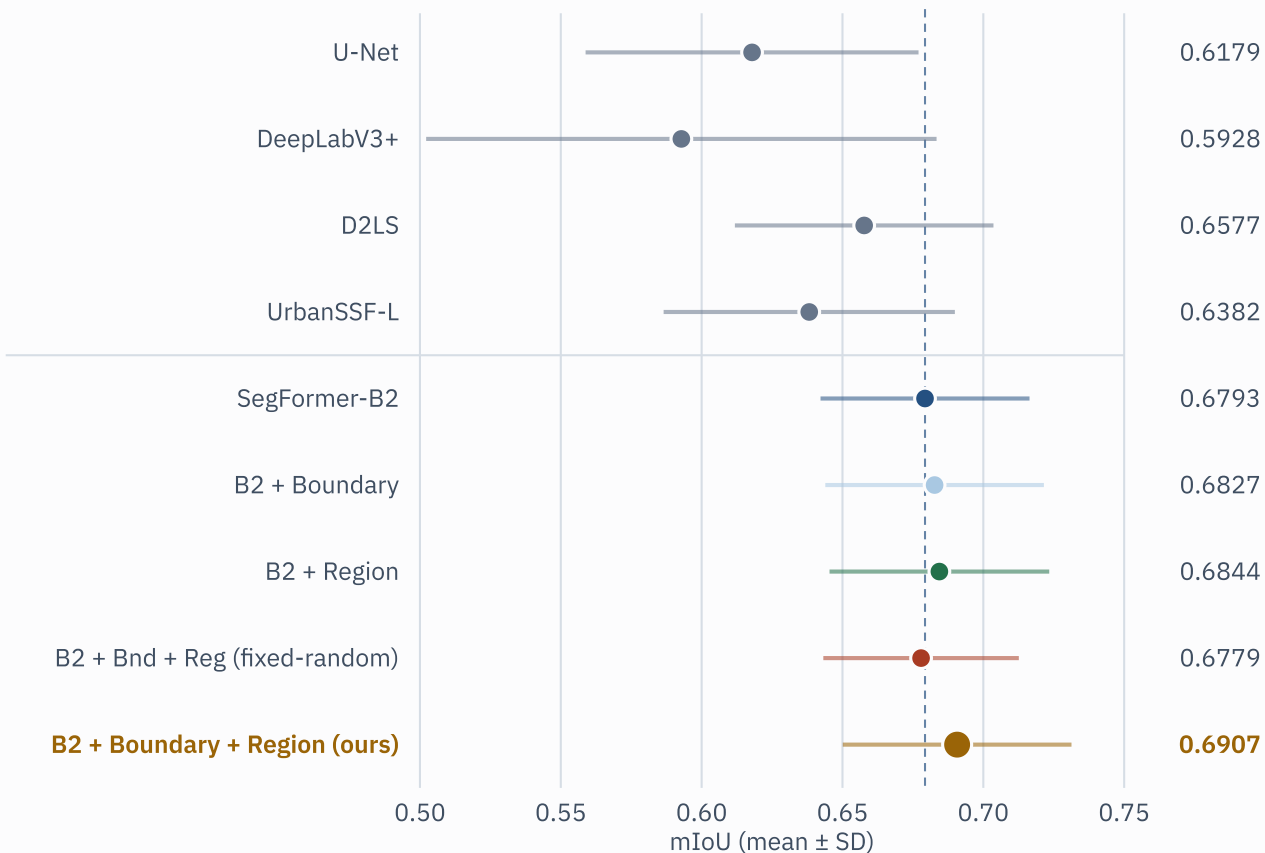
## Region consistency: interiors

$$\tilde{q}_i = \sum_r a_{i,r} \text{softmax}(z_r) \quad \cdot \quad \mathcal{L}_{\text{region}} = \text{KL}(p_i \parallel \tilde{q}_i)$$

- **Eight learnable region prototypes** ( $8 \times 256$ ); cosine similarity of pixel features to prototypes (temperature  $\tau = 0.07$ ) gives soft assignments  $a$
- Region features aggregated by assignment  $\rightarrow$  region classifier  $\rightarrow$  projected back to each pixel as the region-level distribution  $\tilde{q}$
- Pixel predictions  $p$  are pulled towards  $\tilde{q}$  (KL), updating both sides, so predictions within a region stop fragmenting
- **Control:** prototypes sampled and then frozen (fixed-random), isolating prototype trainability

# Main comparison: the joint model leads on all three overlap metrics; each branch alone also helps

mIoU fg\_mIoU mDice



**+0.0114**

mIoU gain over SegFormer-B2 (+1.68%)

**+0.0034 / +0.0051**

mIoU gain of boundary-only / region-only: each helps, the joint model most

**0.6898**

First under assignment A too: mIoU 0.6777 → 0.6898

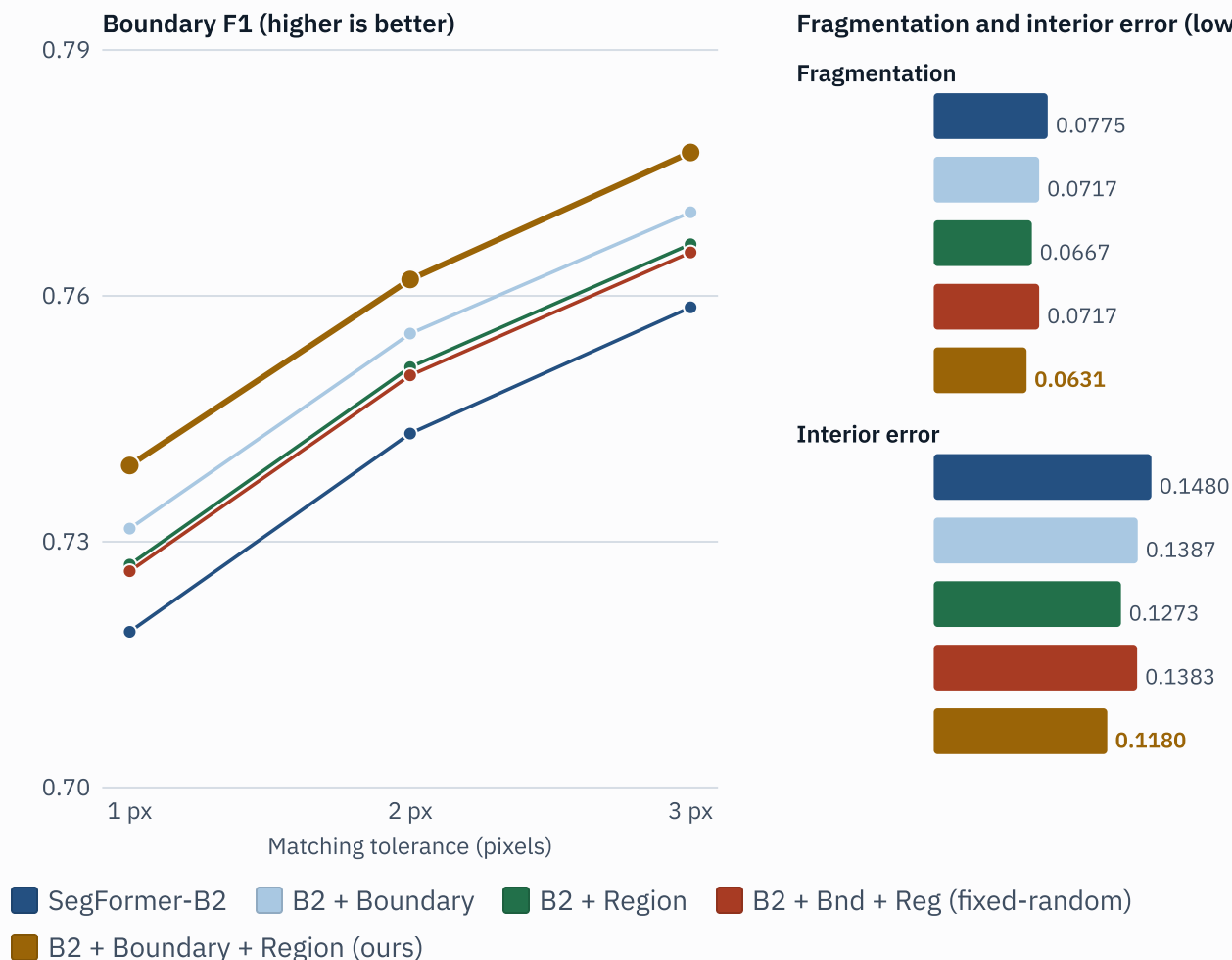
# Highest IoU in all four classes, with the largest margins for buildings and roads

| METHOD                        | BACKGROUND | BUILDING | ROAD   | YARD   | FG_MIOU | MIOU   |
|-------------------------------|------------|----------|--------|--------|---------|--------|
| U-Net                         | 0.6242     | 0.7269   | 0.5516 | 0.5689 | 0.6158  | 0.6179 |
| DeepLabV3+                    | 0.6000     | 0.6885   | 0.5318 | 0.5509 | 0.5904  | 0.5928 |
| D2LS                          | 0.6514     | 0.7852   | 0.5854 | 0.6088 | 0.6598  | 0.6577 |
| UrbanSSF-L                    | 0.6358     | 0.7802   | 0.5562 | 0.5806 | 0.6390  | 0.6382 |
| SegFormer-B2                  | 0.6882     | 0.8020   | 0.6201 | 0.6069 | 0.6763  | 0.6793 |
| B2 + Boundary                 | 0.6938     | 0.8108   | 0.6225 | 0.6037 | 0.6790  | 0.6827 |
| B2 + Region                   | 0.6955     | 0.8105   | 0.6248 | 0.6068 | 0.6807  | 0.6844 |
| B2 + Bnd + Reg (fixed-random) | 0.6833     | 0.8141   | 0.6042 | 0.6100 | 0.6761  | 0.6779 |
| B2 + Boundary + Region (ours) | 0.6960     | 0.8237   | 0.6323 | 0.6108 | 0.6889  | 0.6907 |

Margins over the next-best method: building +0.0096, road +0.0075, yard +0.0008, background +0.0005. Darker cells are higher within their column.

Source: paper Table 7 (class-wise and aggregate IoU over the 15 assignment-B runs, nine methods).

# Structural quality: sharper boundaries, fewer fragments, fewer interior errors; the two branches complement each other



**0.7190 → 0.7393**

Boundary F1 at 1-pixel tolerance; also highest at 2 and 3 pixels

**-18.6%**

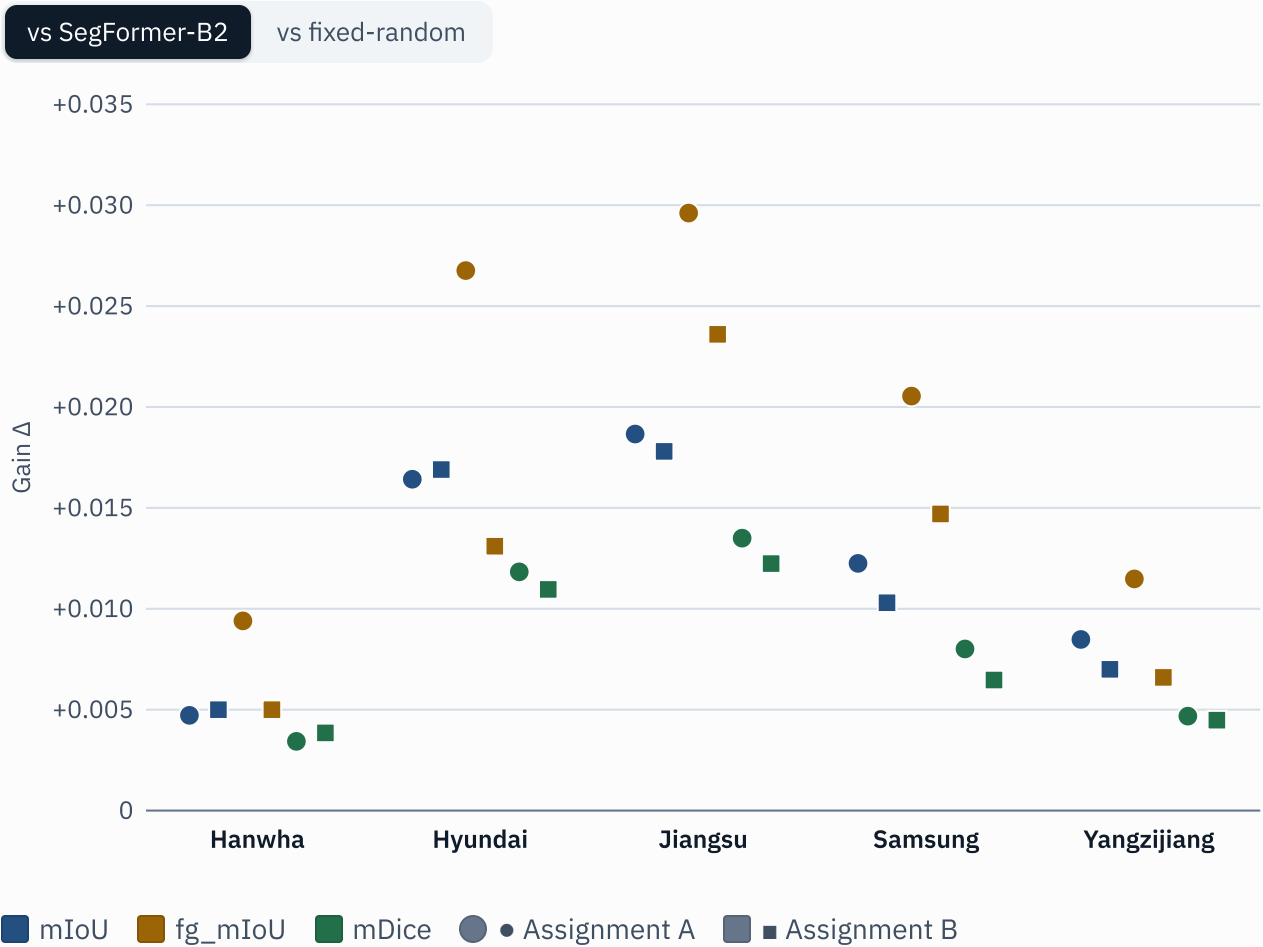
Small-component ratio (fragmentation) 0.0775 → 0.0631

**-20.3%**

Interior error rate 0.1480 → 0.1180; yard 0.158 → 0.116

Boundary-only helps boundary F1 more; region-only helps fragmentation and interior errors more; the joint model gets both.

# Site by site: gains are positive for every held-out yard under both validation assignments



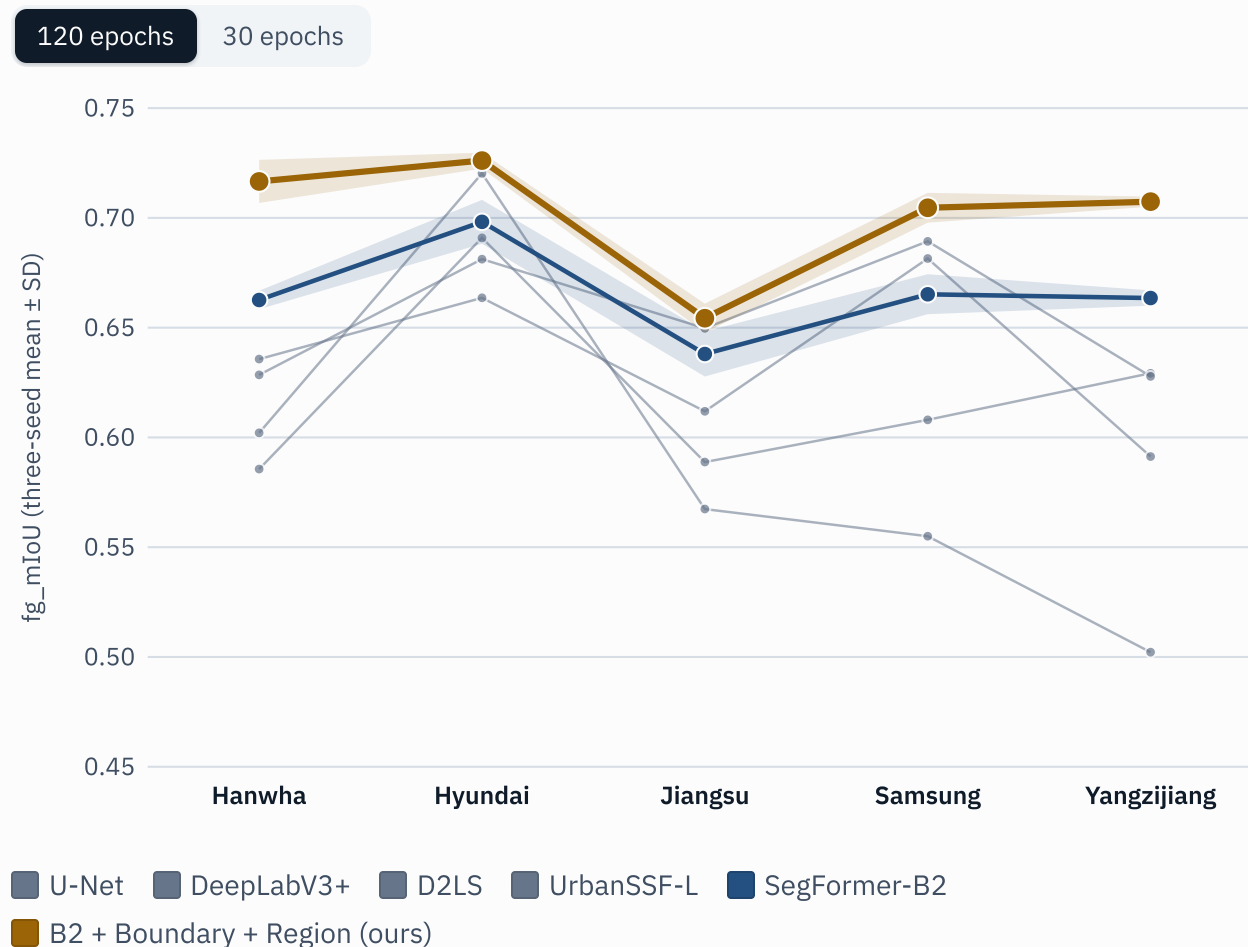
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site-assignment comparisons, all positive on all three metrics

|         | SITE-LEVEL MEAN $\Delta$ | 95% INTERVAL       | POSITIVE YARDS | EFFECT SIZE $D_z$ |
|---------|--------------------------|--------------------|----------------|-------------------|
| mIoU    | +0.01175                 | [0.00730, 0.01621] | 5/5            | 2.06              |
| fg_mIoU | +0.01608                 | [0.01002, 0.02214] | 5/5            | 2.01              |
| mDice   | +0.00795                 | [0.00474, 0.01115] | 5/5            | 1.95              |

Sources: paper Tables 10–12. Each point is one test yard's three-seed mean gain under one assignment. Site-level pairing averages both assignments and three seeds per yard, giving five paired differences; the 95% interval is a percentile interval over all 3,125 resamples. With  $n = 5$  all positive, the exact two-sided Wilcoxon  $p = 0.0625$ , the smallest value five pairs can reach.

# Longer training: at 120 epochs the advantage holds, above the baseline on all five yards



**+0.0163 – +0.0540**

fg\_mIoU gain over SegFormer-B2 on the five yards: +0.0163 to +0.0540

**0.6942**

120 epochs: mIoU 0.6782 → 0.6942

**0.7018**

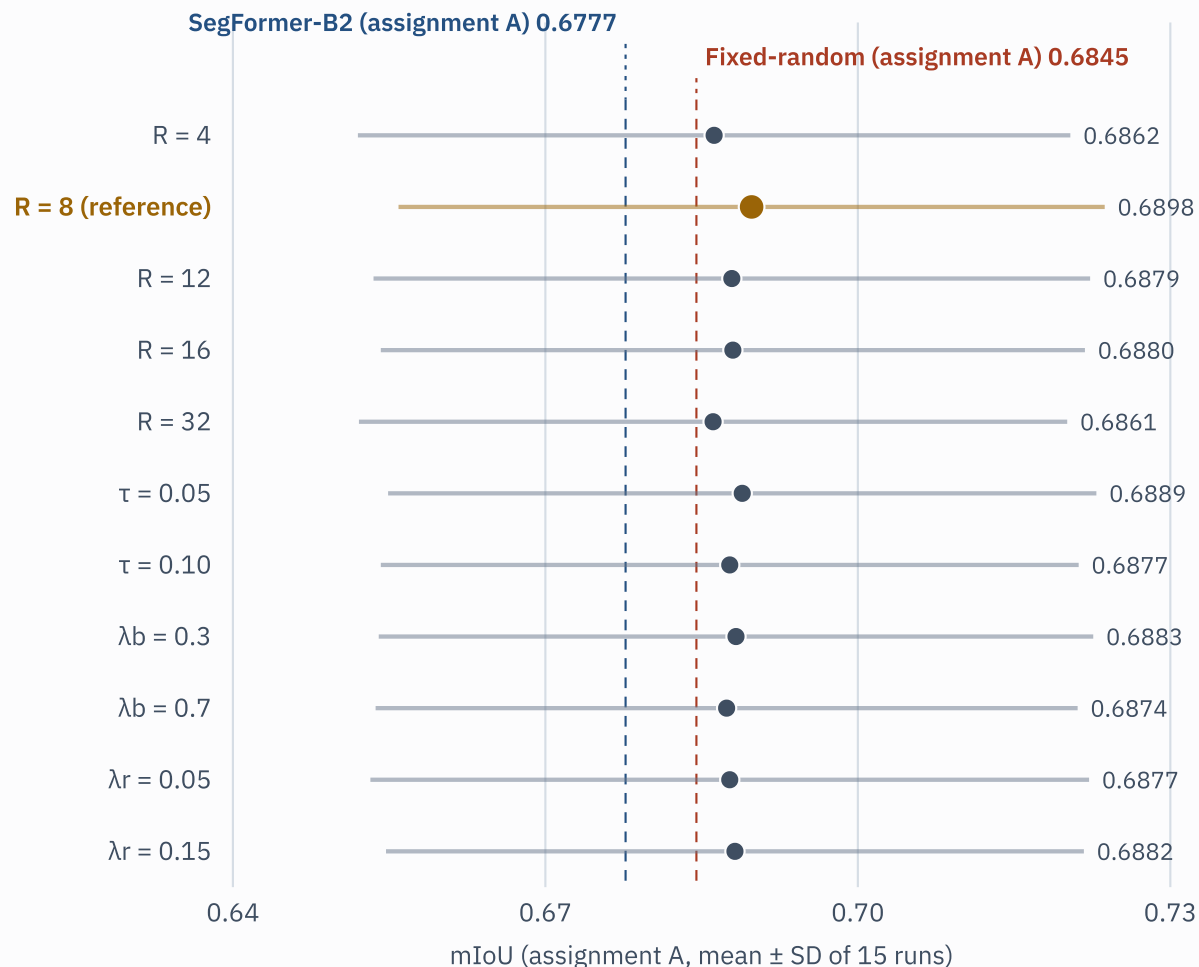
120 epochs: fg\_mIoU 0.6655 → 0.7018

**0.8103**

120 epochs: mDice 0.8021 → 0.8103

Sources: paper Tables 15, 17, 18 and §5.7. At 120 epochs each method uses its own reference learning rate and a cosine schedule; across 0.5×, 1× and 2× learning rates the joint model ranks first in all three metrics at each multiplier, and its lowest mean over the grid exceeds every comparator's highest.

# The prototypes do learn: learnable beats fixed-random, and nearby settings stay robust



**+0.0091** mIoU

Site-level mean gain over fixed-random prototypes, 95% interval [0.0062, 0.0113], positive on 5/5 yards

**0.512 → 0.802**

Per-position maximum assignment: learned prototypes assign more decisively (normalised entropy 0.641 → 0.312)

**7.56–7.95** / 8

Effective number of prototypes: all eight are used overall, no collapse

**All 11 one-factor settings on the left beat both SegFormer-B2 and the fixed-random control.**

# Free at deployment: the inference model is exactly SegFormer-B2

## 27.35 M

deployment parameters (same as baseline)

## 92.1 G

FLOPs (same as baseline)

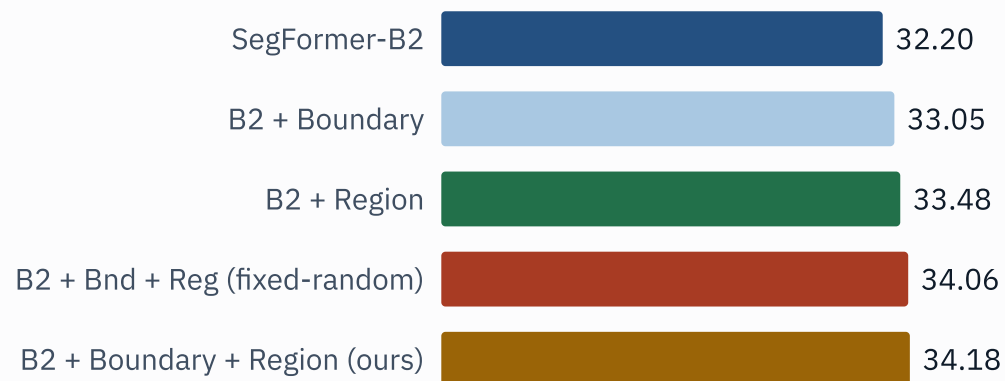
## 31.10 ms

latency per image; baseline 31.14 ms

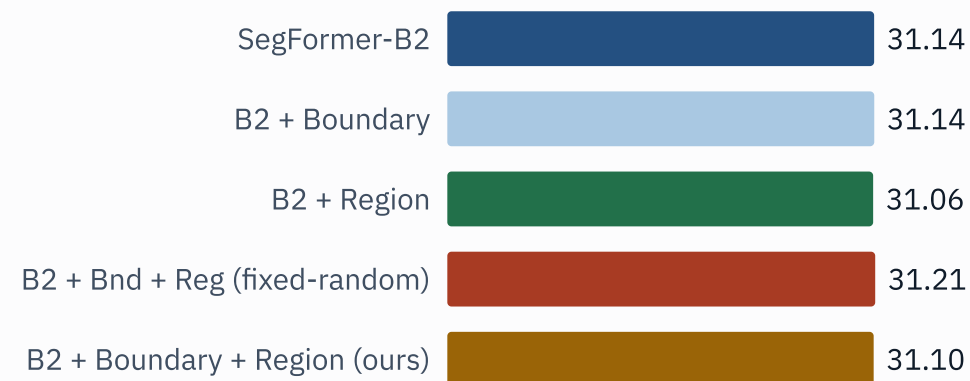
## +6.1%

training per epoch 32.20 → 34.18 s;  
364,165 auxiliary parameters exist only in training

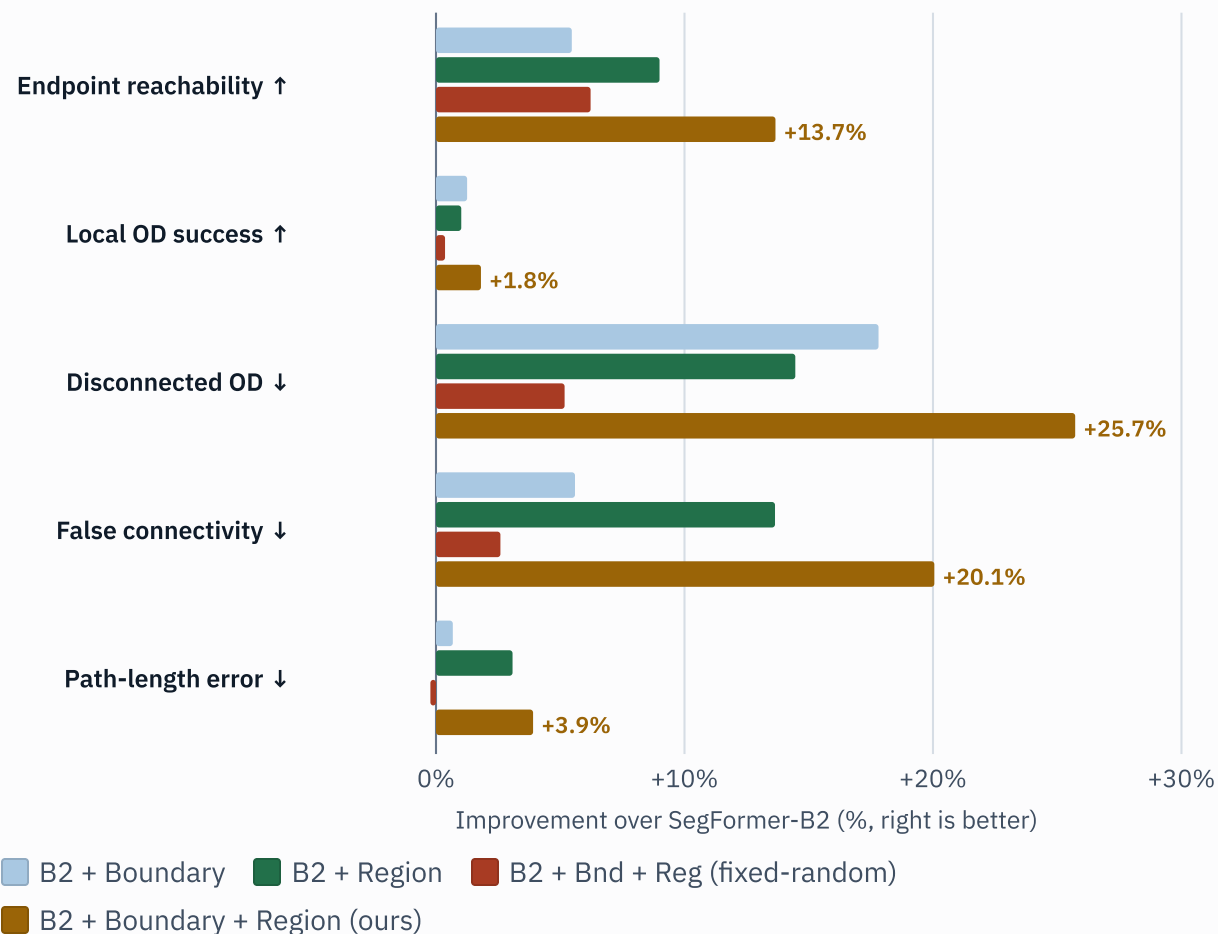
### Training (s / epoch)



### Inference latency (ms / image)



# Towards road networks: better local connectivity, while long-range reachability still needs topology repair



**6.57% → 4.88%**

Local path failures (origin–destination pairs 4.8–19.2 m apart within one connected road)

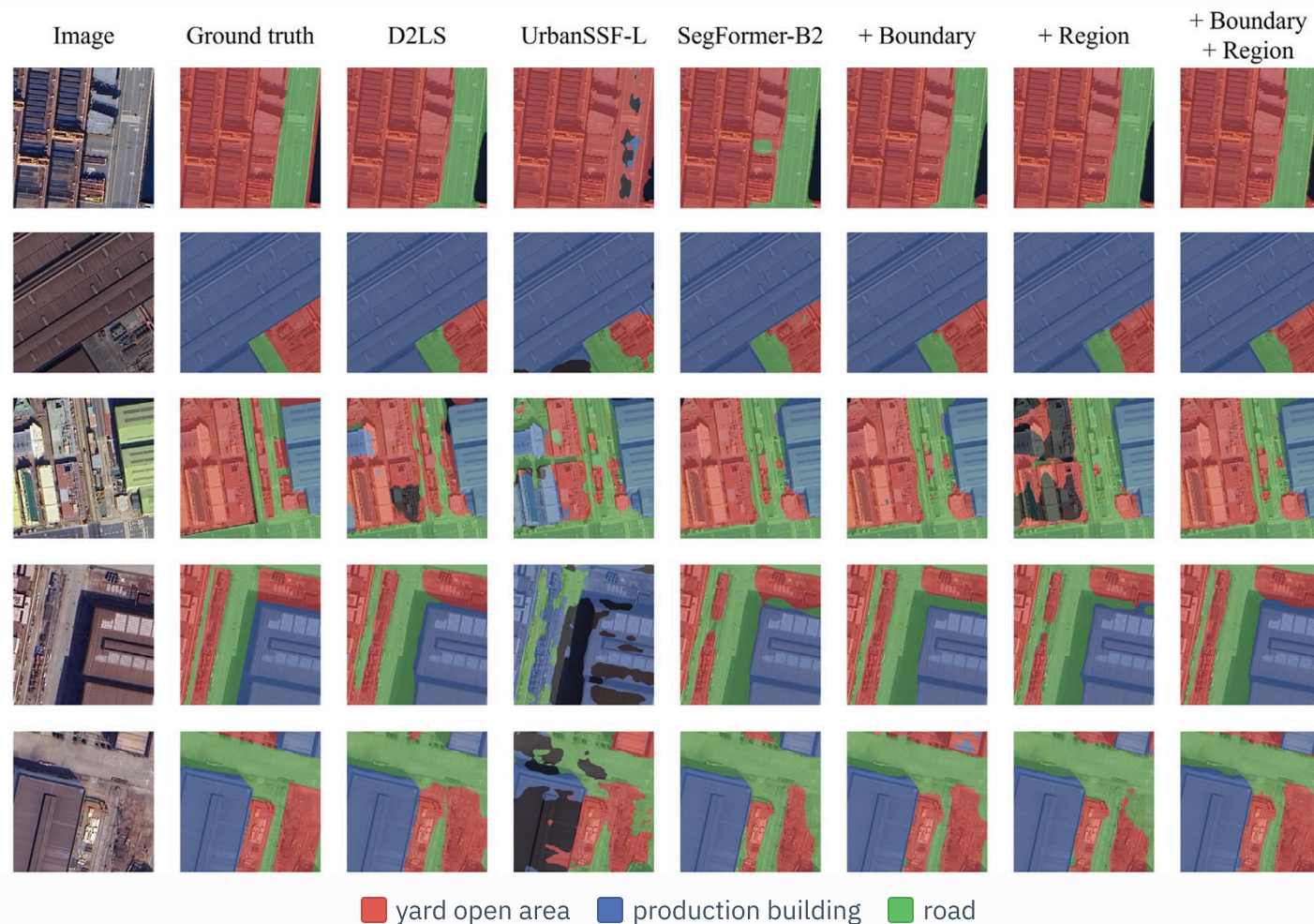
**0.0733 → 0.0586**

False connectivity: separate roads wrongly joined

**0.2704**

Long-range endpoint reachability ( $\geq 38.4$  m): still low, so transport planning needs topology repair and GIS vectorisation first

# Qualitative comparison on the five held-out yards: clearer transitions, fewer interior fragments



Source: paper Fig. 10 (30-epoch models; imagery Google, Airbus). Columns: image, ground truth, D2LS, UrbanSSF-L, SegFormer-B2, +Boundary, +Region, +Boundary+Region (ours).

# Local improvements and the errors that remain

Gains: (a) road–yard interface (b) building boundary (c) yard interior



Residual errors: (a) road–yard confusion (b) shadow or occlusion (c) thin connection broken



# Conclusions: complementary training constraints make cross-yard maps more accurate and cleaner at no deployment cost

1

## A cross-yard protocol

Five held-out yards, two validation assignments, three seeds

2

## Two training constraints

Morphological boundaries + prototype region consistency, removed at inference

3

## Controlled ablations

Separating boundary, region and prototype trainability

4

## Complementary structural gains

Interface and interior errors both fall; deployment graph unchanged

## Its place in the thesis

It supplies semantic base maps that can speed up layout extraction for chapter 6's many yards; road topology is still traced as centrelines, with segmentation as support and citation.

## Next

External tests across regions, sensors and seasons; whether topology-corrected maps improve route planning.

[Thesis briefing →](#)

[Code →](#)

**Boundaries fix the transitions, regions fix the interiors: complementary, and free at deployment.**

**Thank you.**

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